

Data Science Report: Employee Efficiency Analysis

Introduction

This report presents the findings from the analysis of employee work times across various tasks and projects. The data was sourced from a dataset containing detailed records of tasks, work durations, and other relevant information. The primary objective of this study was to uncover insights related to employee efficiency, and how it fluctuates across various tasks, clients, and other factors.

Data Overview

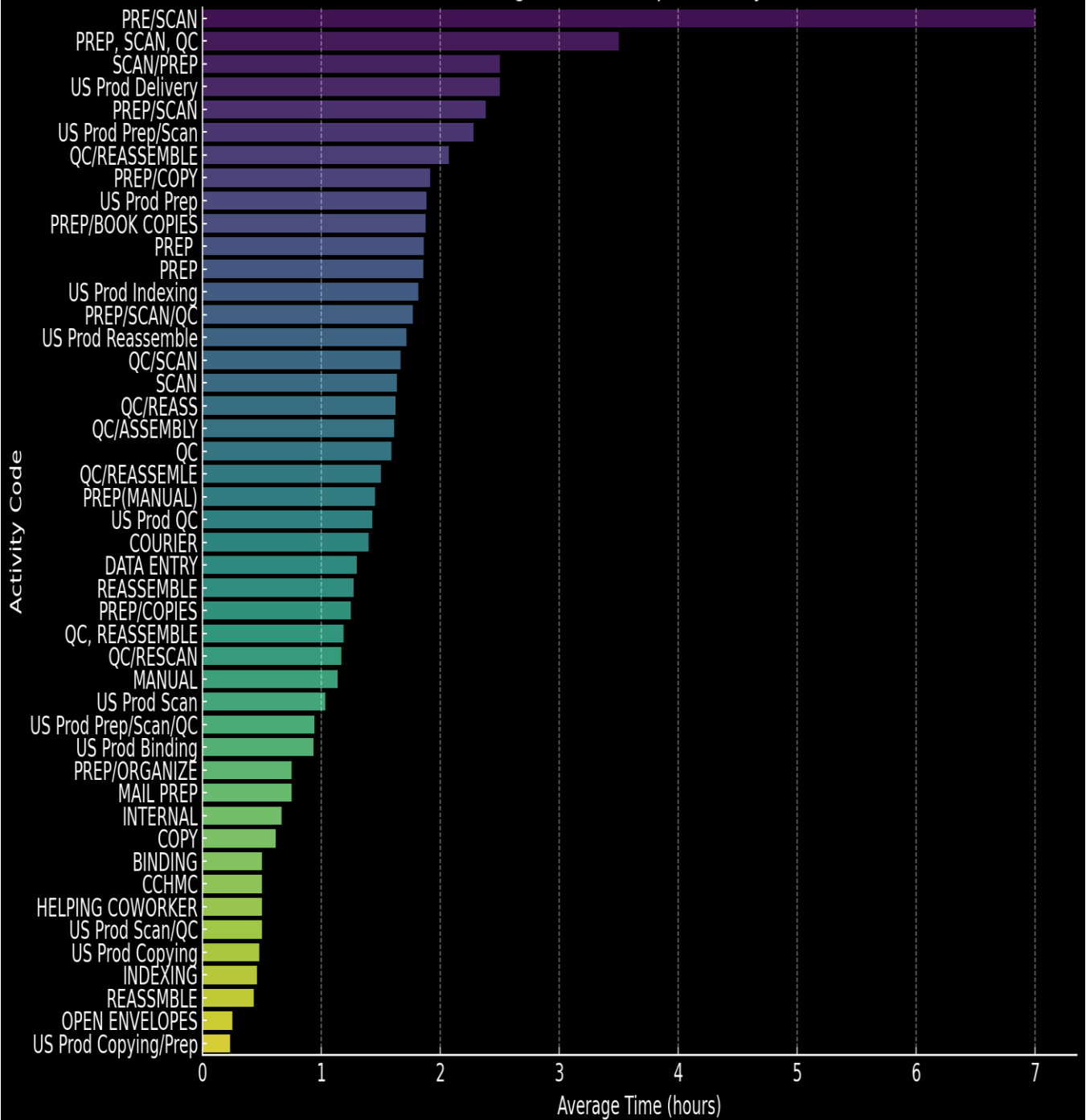
The dataset under scrutiny contains task records completed by employees, each of which details the client, project name, activity code, staff member, start and end times, and the total hours spent on each task. Prior to analysis, the dataset underwent a cleaning process to handle missing values and anomalies.

Key Findings

Task Completion Time

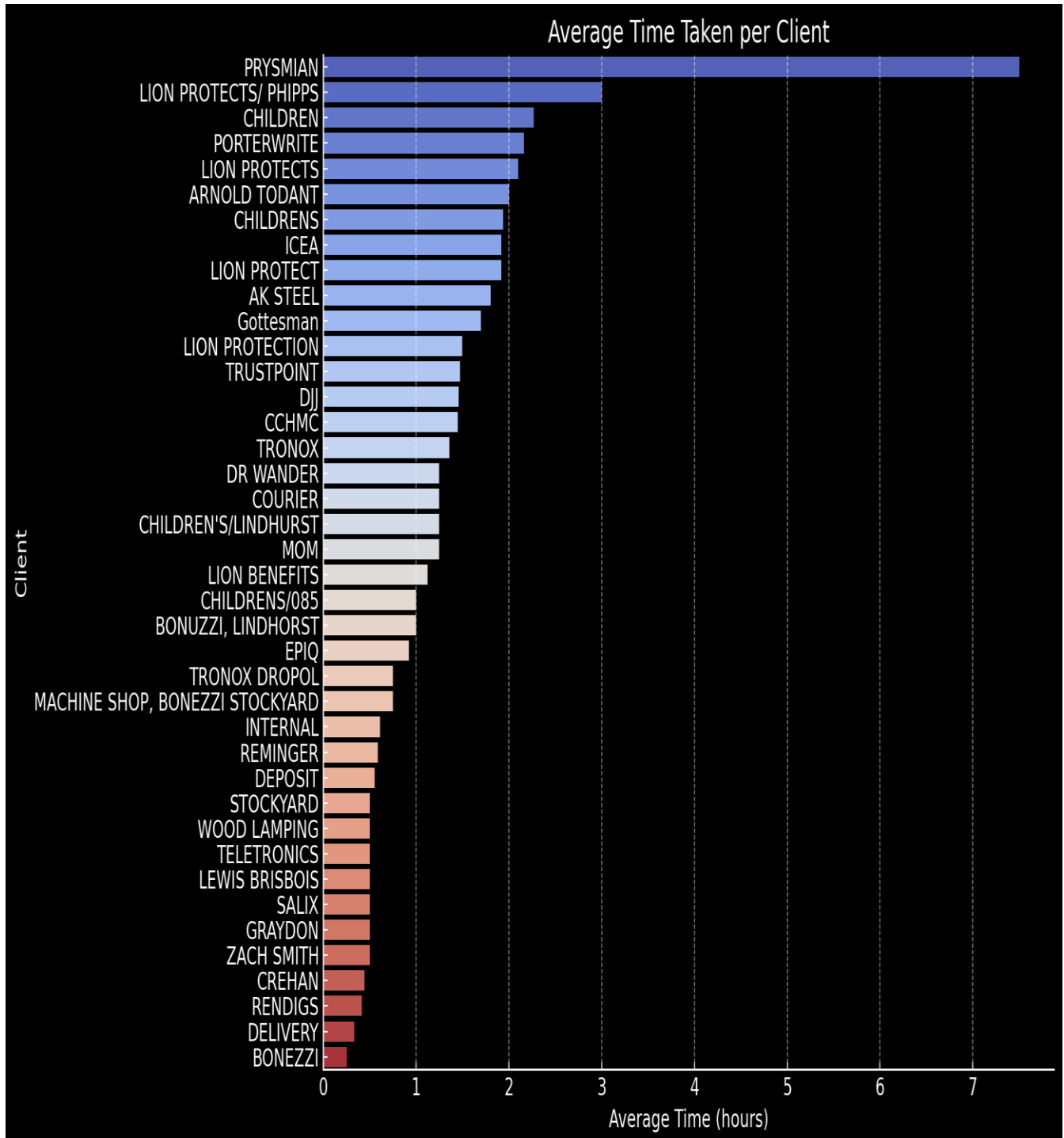
The average time taken to complete a task by the production team in the USA was found to be around 1.64 hours, with a standard deviation of 1.37 hours. This indicates a relatively wide spread in the task completion times.

Average Time Taken per Activity Code



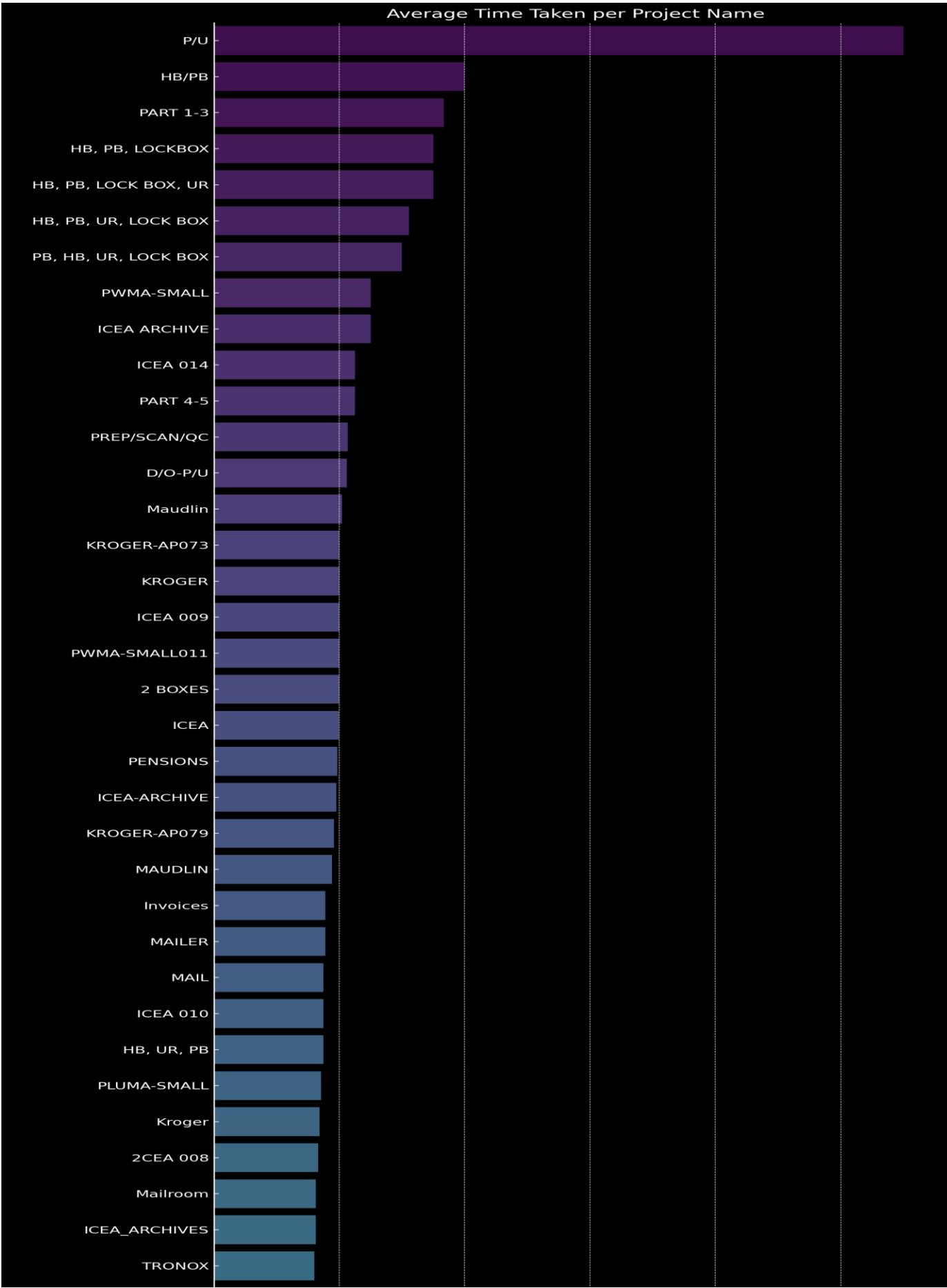
Client-Specific Analysis

Similar to the project-specific analysis, the average time taken to complete tasks also varied across different clients. Tasks related to "AK STEEL" had an average completion time of 1.8 hours, while those associated with "ARNOLD TODANT" took longer, with an average of 2hrs. While "PRYSMIAN" took the longest time of more than 7hrs, "BONEZZI" took less than 30 minutes

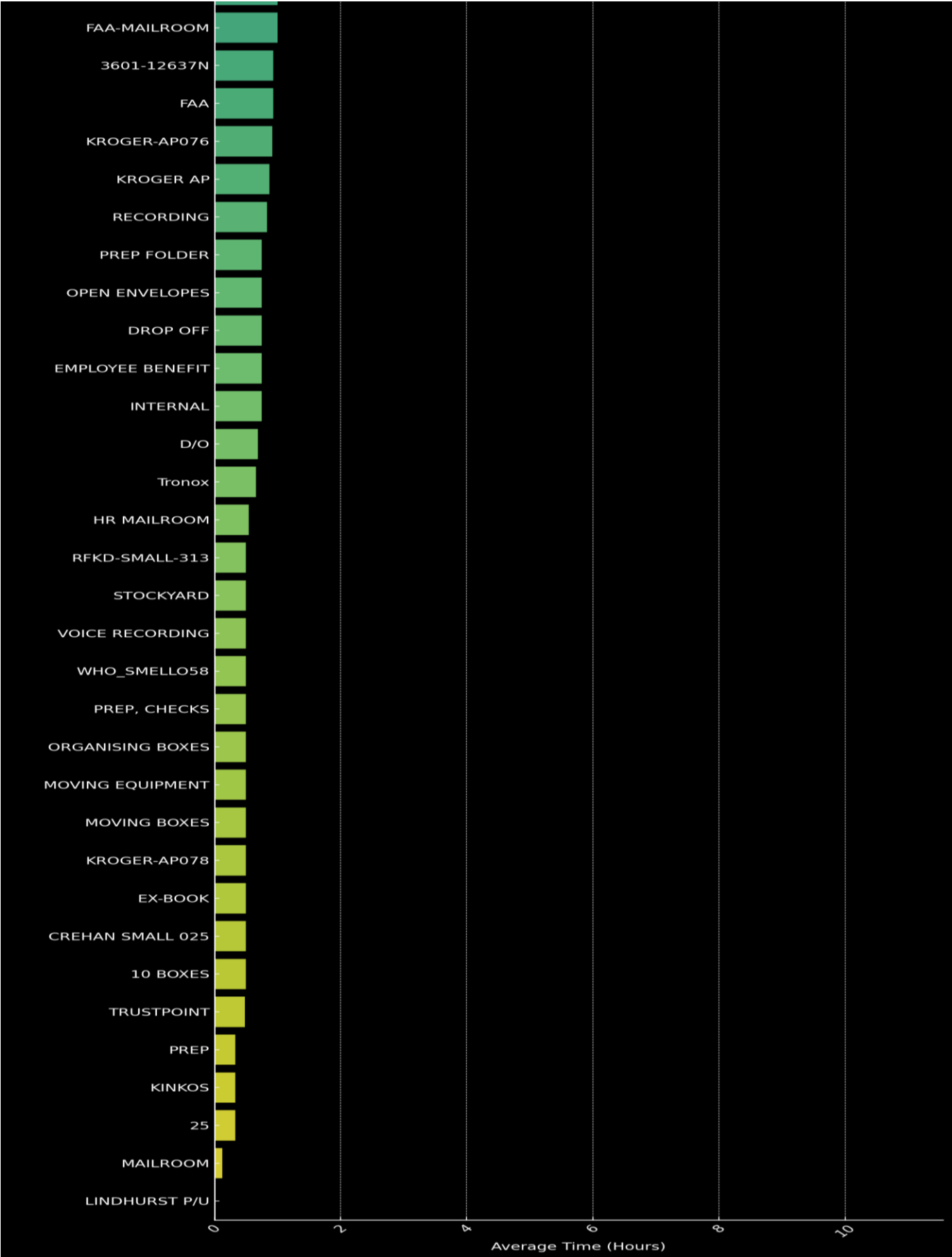


Project-Specific Analysis

The data revealed significant variations in the average time taken to complete tasks across different projects. For example, tasks associated with Project "0" had an average completion time of 1.82 hours, while those related to Project "25" were completed much faster, with an average time of 0.33 hours.



Project Name	BENEFITS					
	HB,PB, UR					
	HB, PB, UR					
	MAILER(TRONOX)					
	PWMA-SMALL013					
	BOX 001					
	ARCHIVE					
	Gottesman					
	6A-SMALL					
	ICEA_ARCHIVE					
	KROGER-AP					
	KROGER-AP081					
	ICEA-ACHIVE					
	AK STEEL					
	HB, PB					
	PWMA-SMALL012					
	DJJ/CCHMC					
	TRONOX(MAILER)					
	P/O					
	KROGER-AP077					
	PREP/D/O					
	KROGER_AP085					
	KROGER-AP080					
	KROGER-AP083					
	KROGER-APO83					
	DROPOFF					
	FAA-MAILROOM					
	3601-12637N					

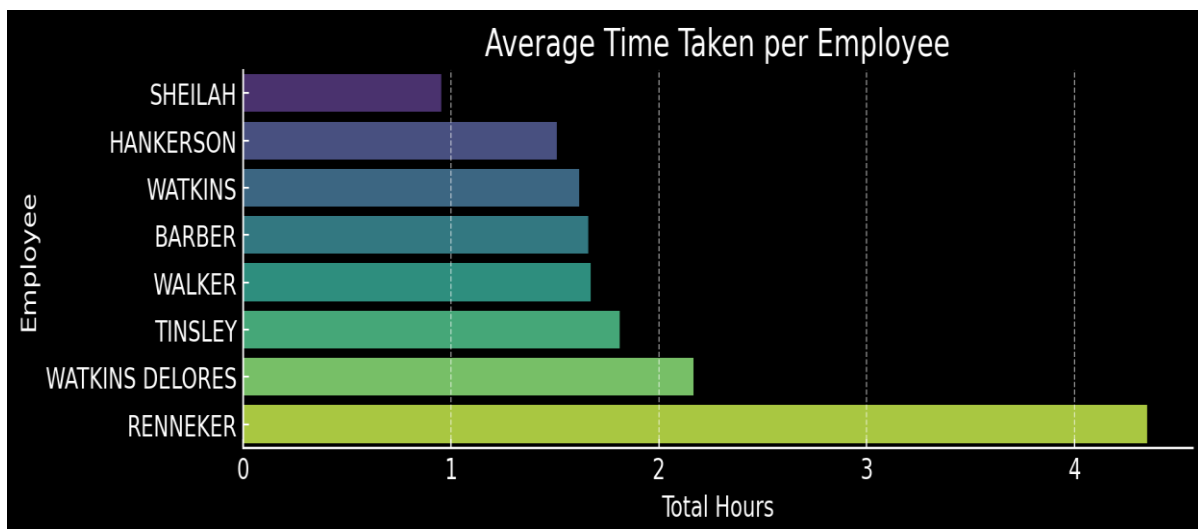


Employee Efficiency

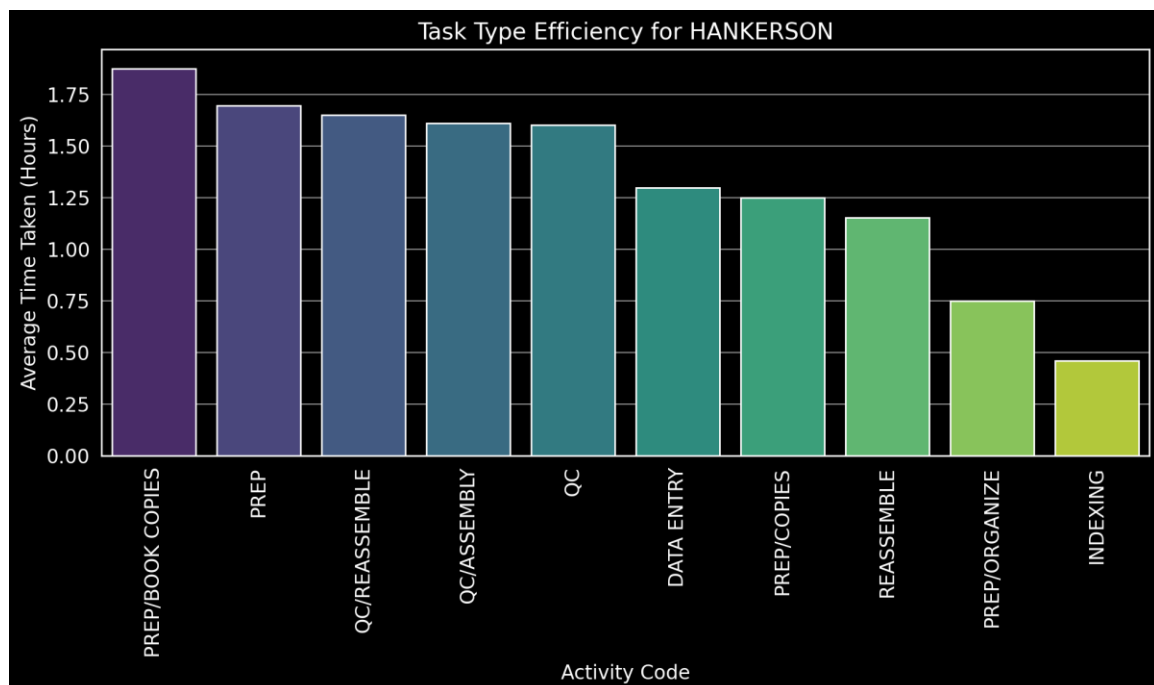
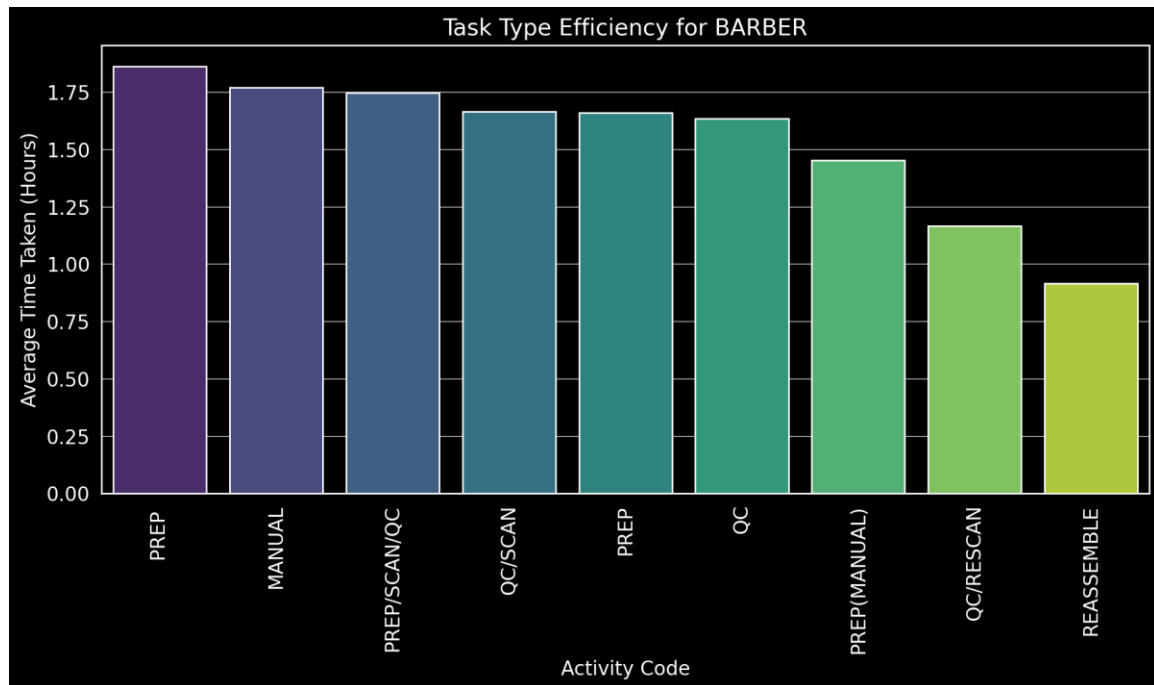
A comparison of the time taken to complete each activity by different employees revealed noticeable differences in efficiency. For example, Employee A's average task completion time was different from that of Employee B. Factors contributing to these differences may include the nature of the tasks, employee skills and experience, among others.

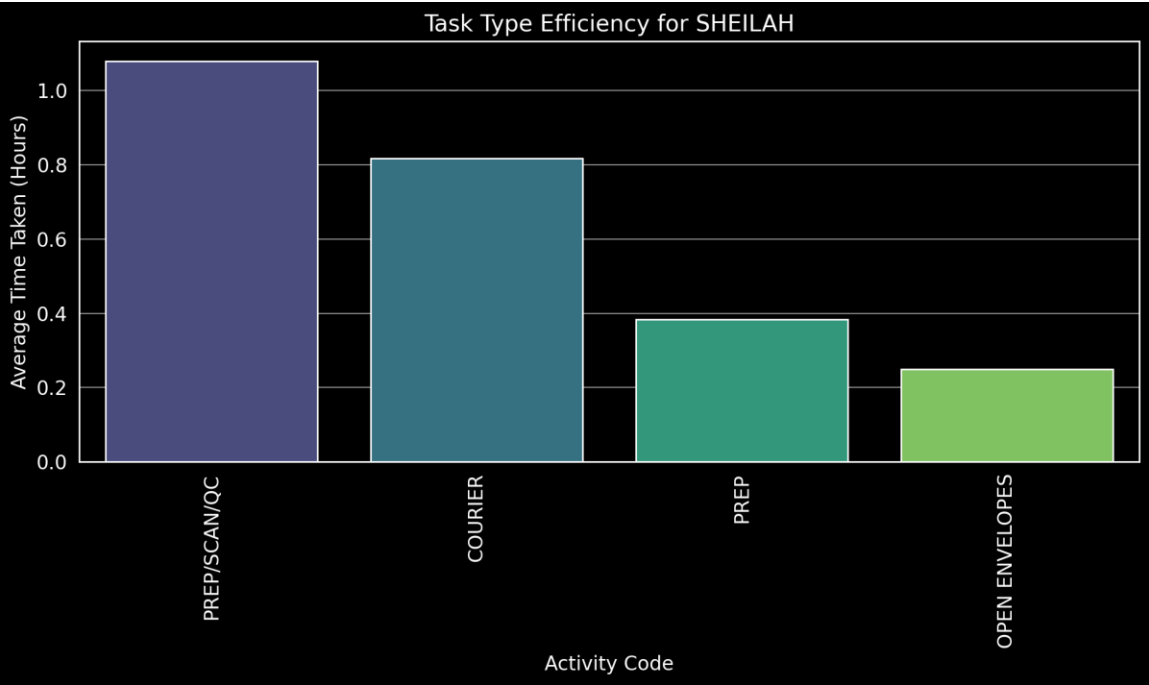
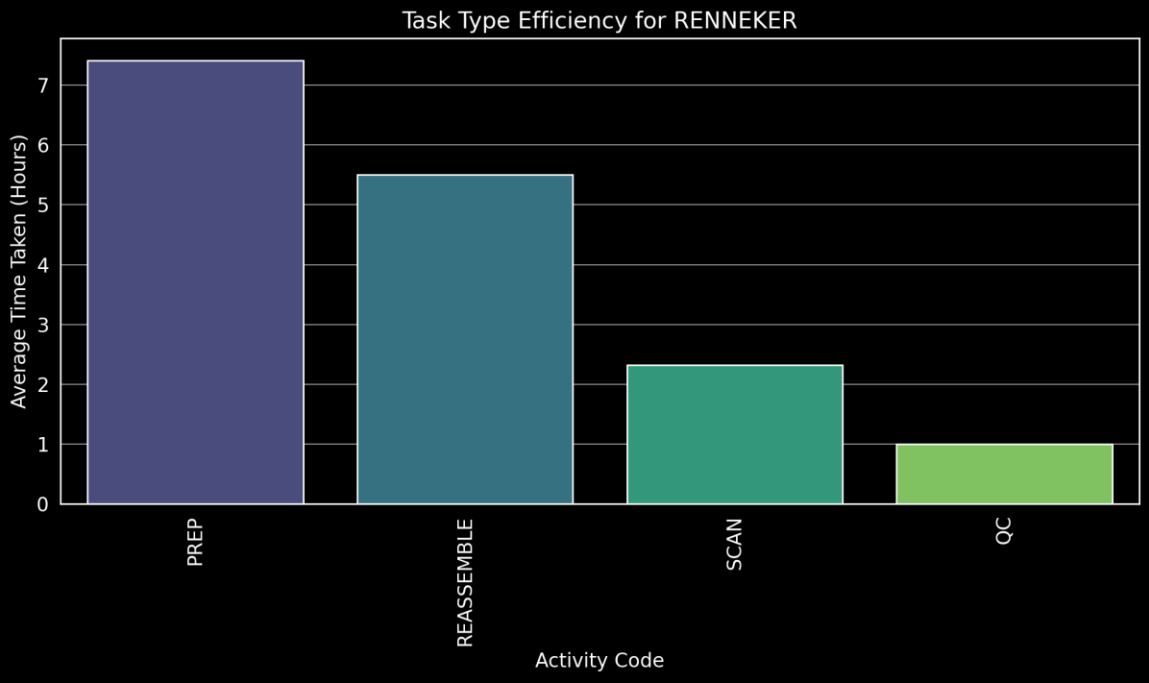
Based on the cleaned data, here is the average time taken to complete tasks for each staff member:

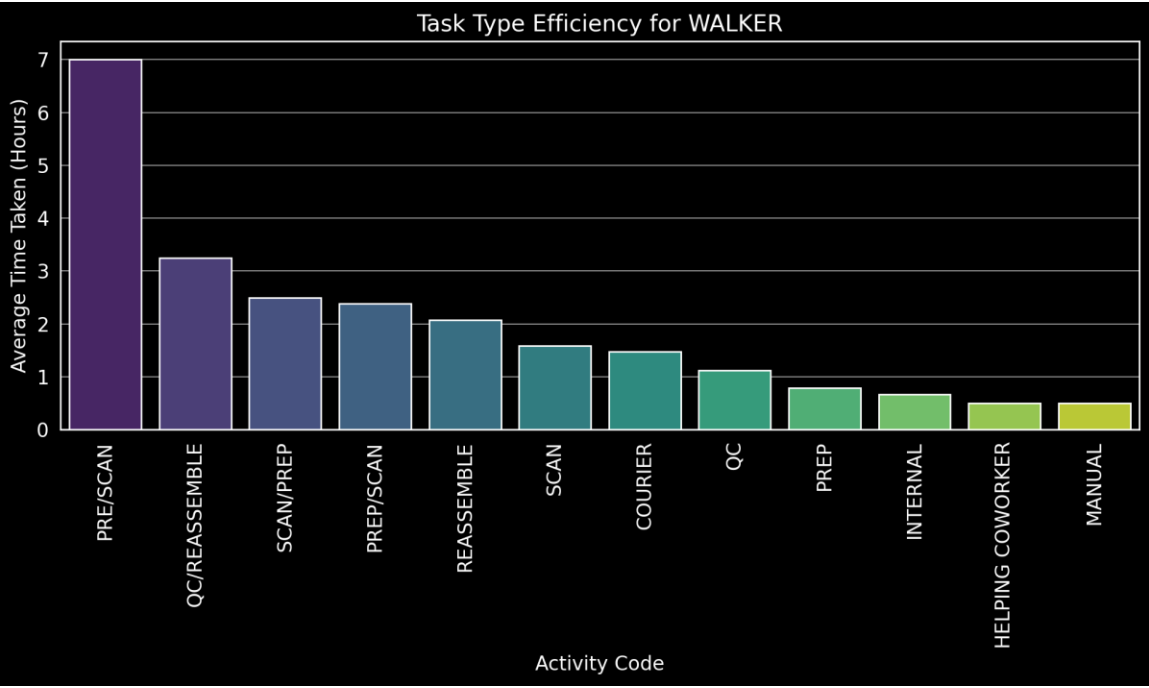
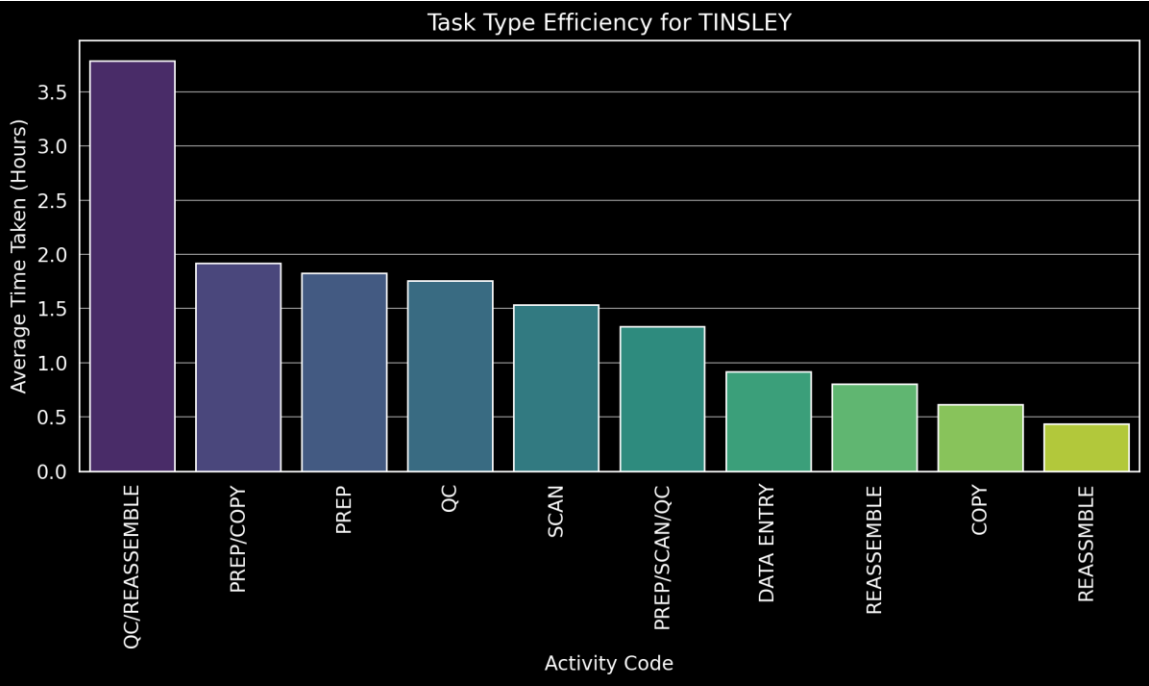
STAFF	TOTAL HOURS
RENNEKER	4.35
WATKINS DELORES	2.17
TINSLEY	1.81
WALKER	1.67
BARBER	1.66
WATKINS	1.62
Unknown	1.59
HANKERSON	1.51
SHEILAH	0.96

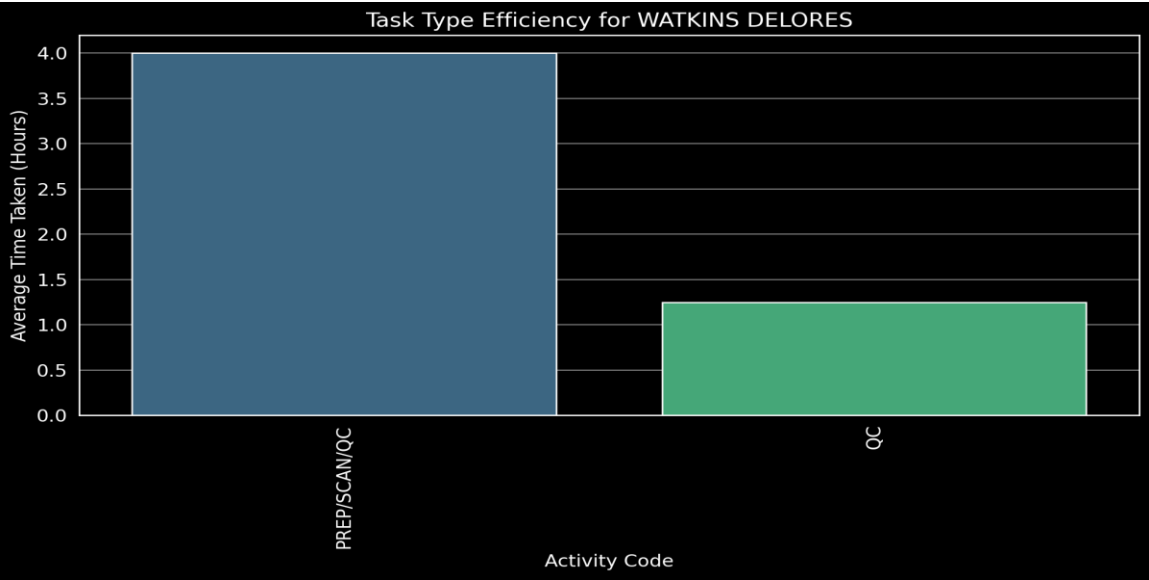
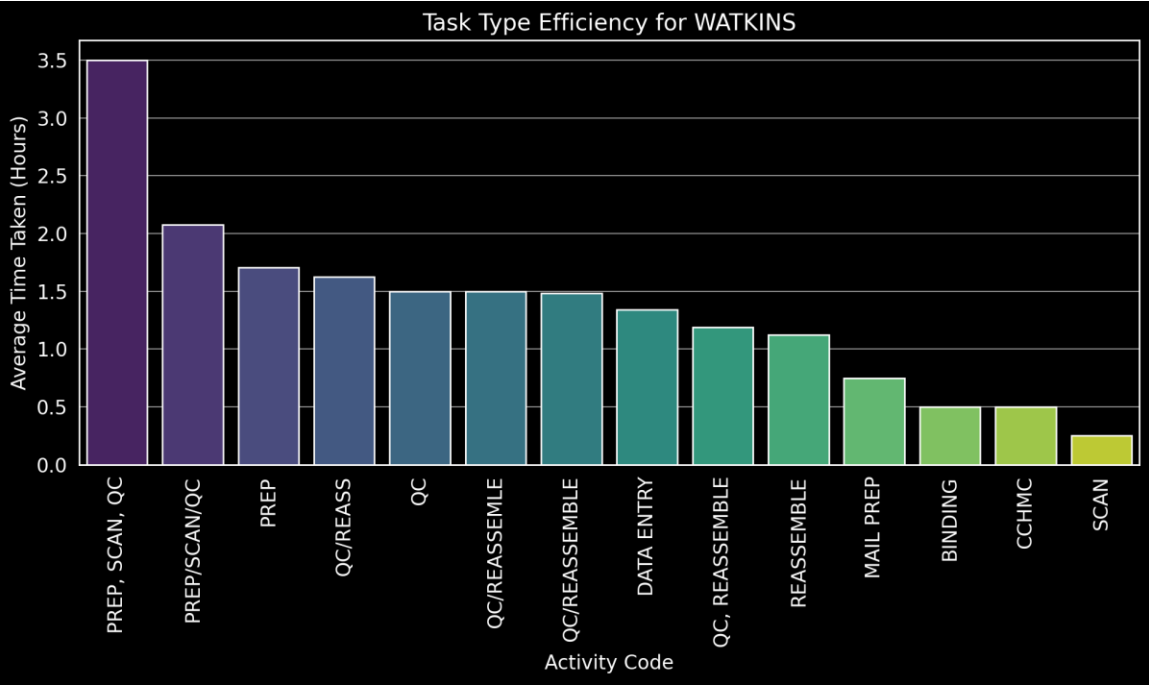


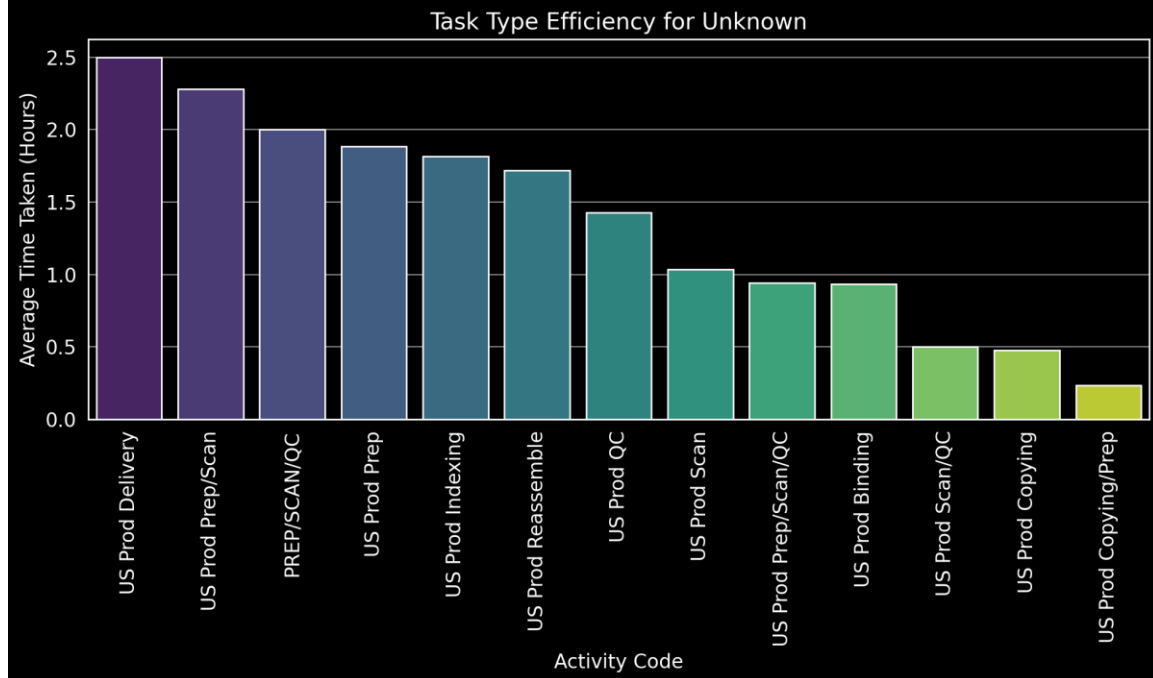
From the table, we can observe noticeable differences in the average time taken by different staff members. For instance, the staff member 'RENNEKER' takes the longest average time (about 4.35 hours), while 'SHEILAH' takes the shortest average time (about 0.96 hours).











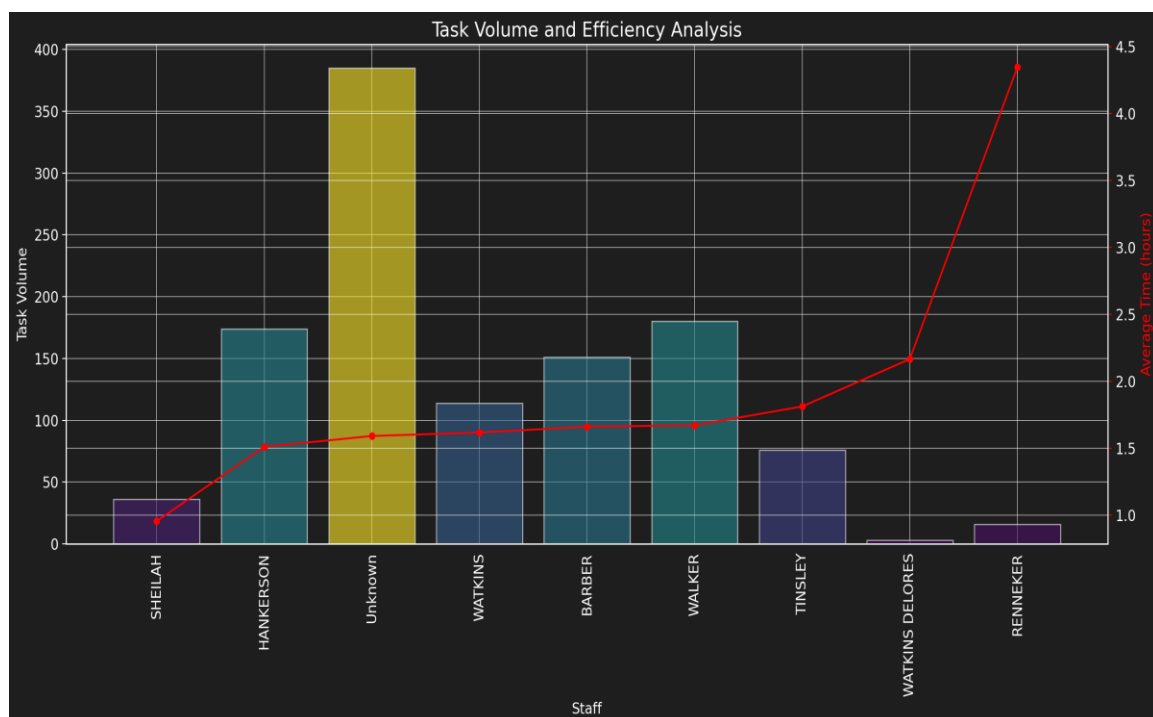
Variations Across Tasks: The charts display noticeable variations in the time taken for different activity codes within individual employees. This suggests that certain tasks may require more time to complete due to complexity, skill requirements, or other factors.

Employee-Specific Efficiency: Some employees may excel in specific tasks and take less time to complete them, while others may take longer for the same tasks. Understanding these individual efficiencies can help in assigning tasks based on employee strengths.

Potential for Optimization: By identifying the tasks that typically take longer and understanding why, strategies can be developed to optimize these tasks, whether through additional training or process improvements

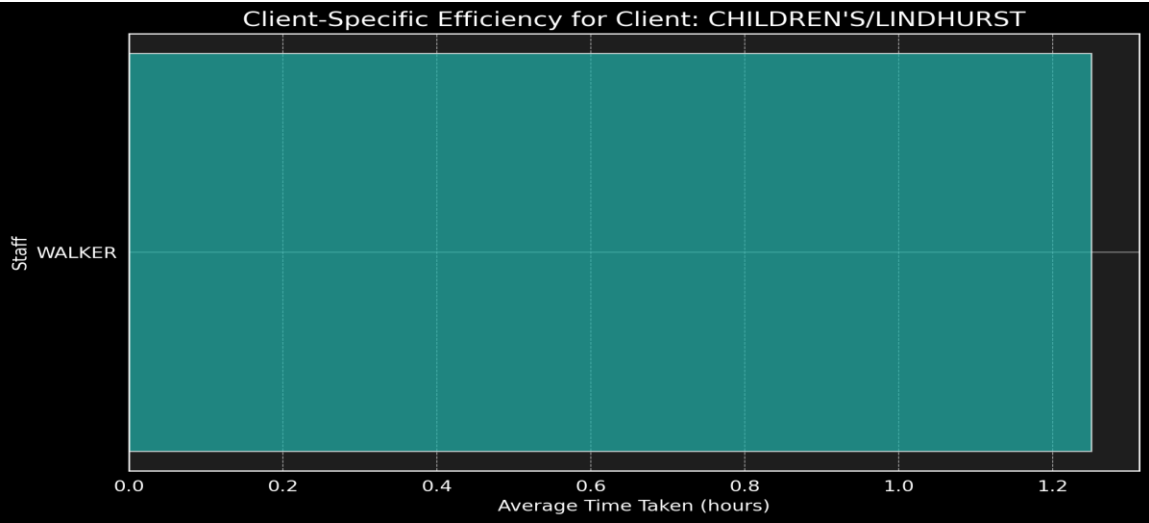
Task Volume and Employee Efficiency Analysis

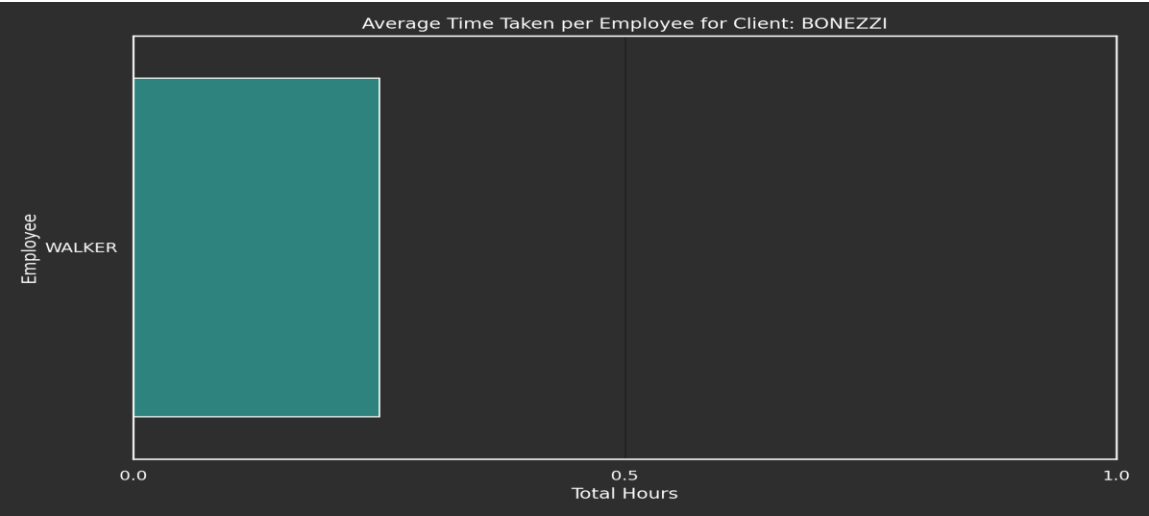
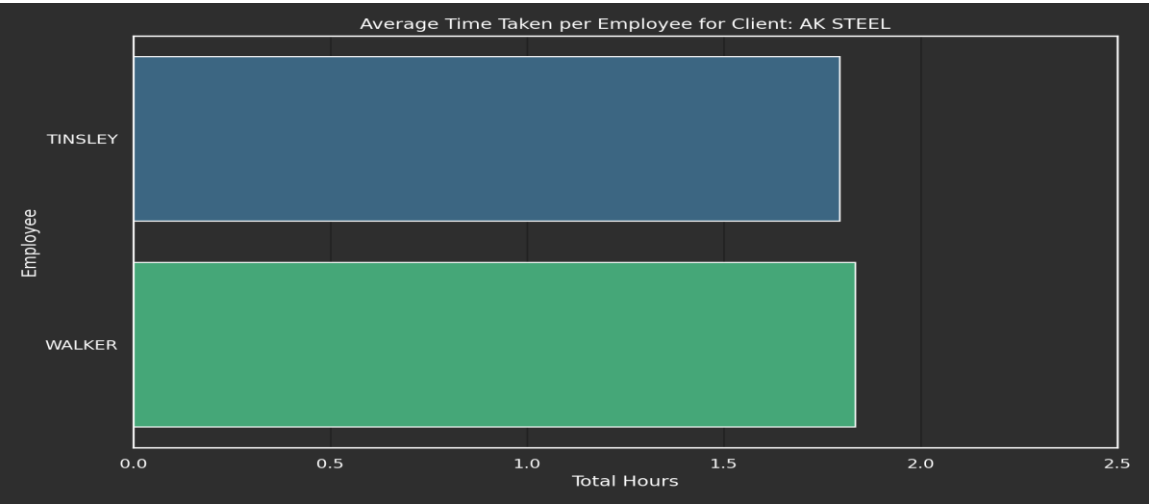
The bar plot represents the task volume, i.e., the total number of tasks handled by each employee. **The line chart**(in red) represents the average time taken to complete their tasks.

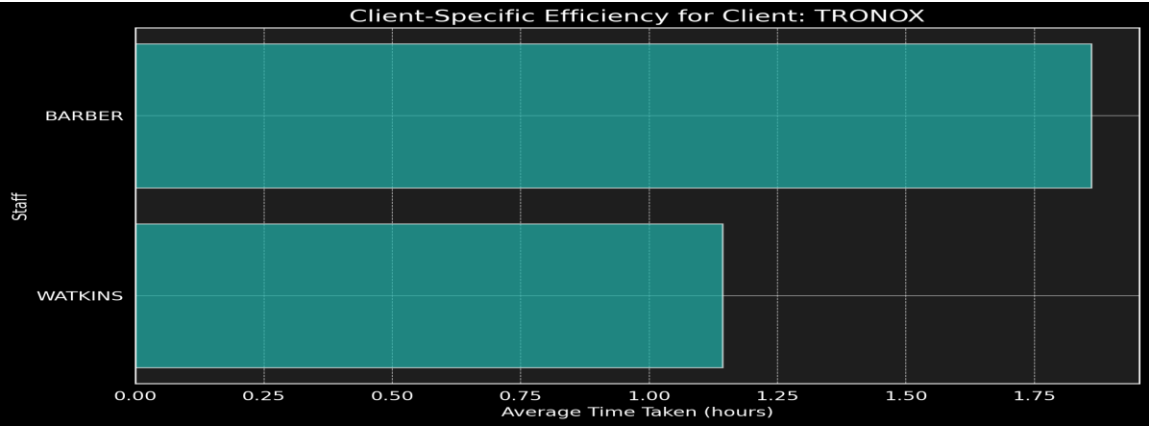
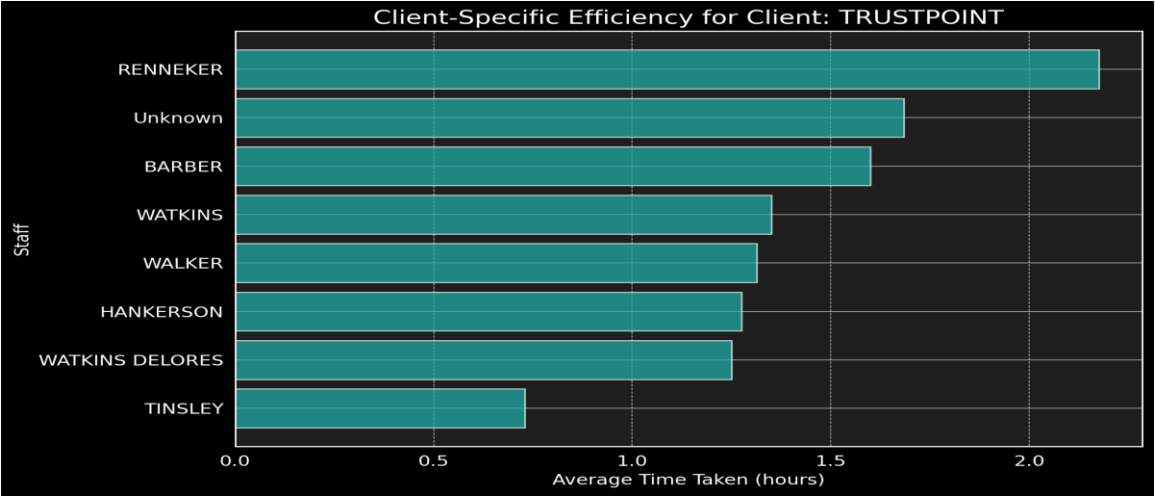


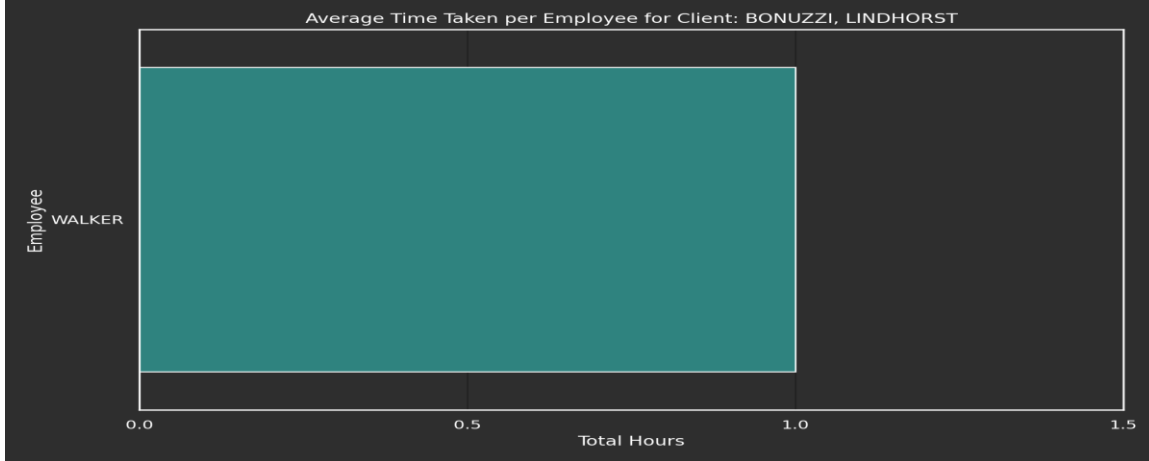
Client Specific Employee Efficiency.

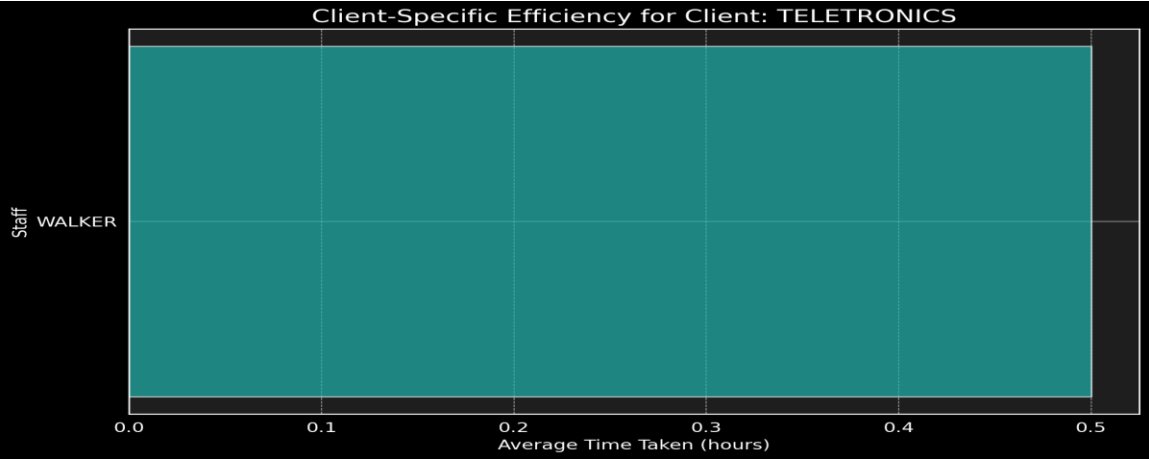


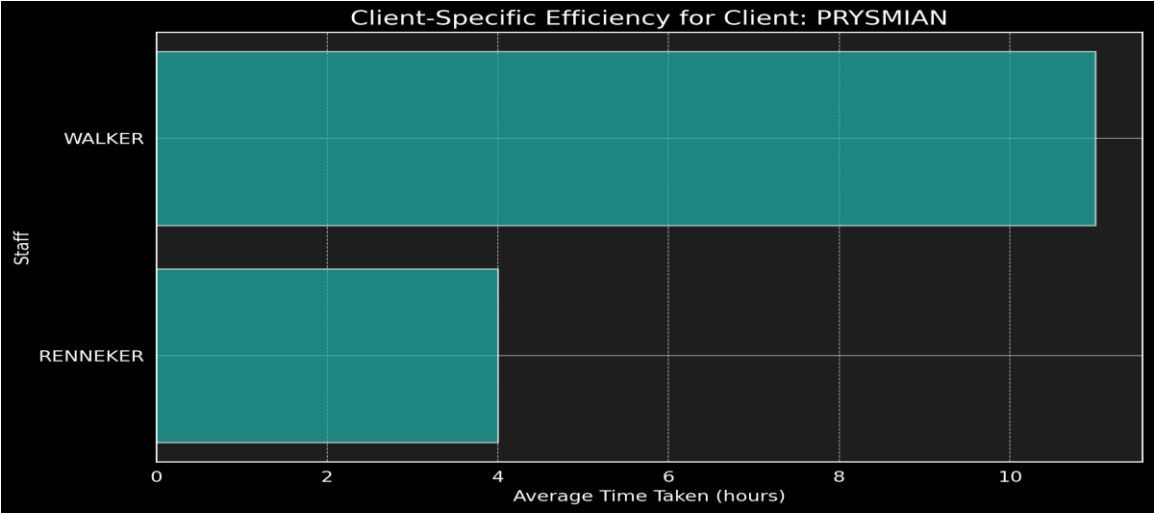


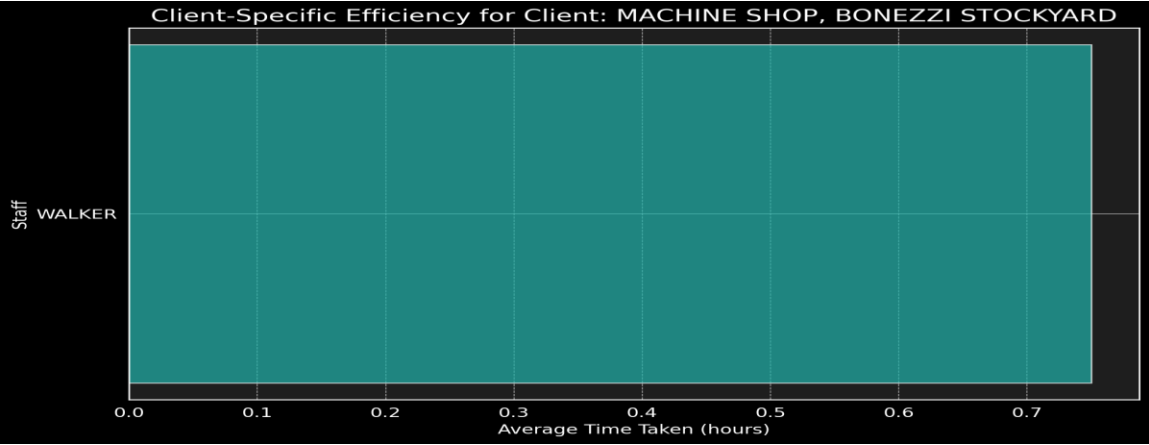
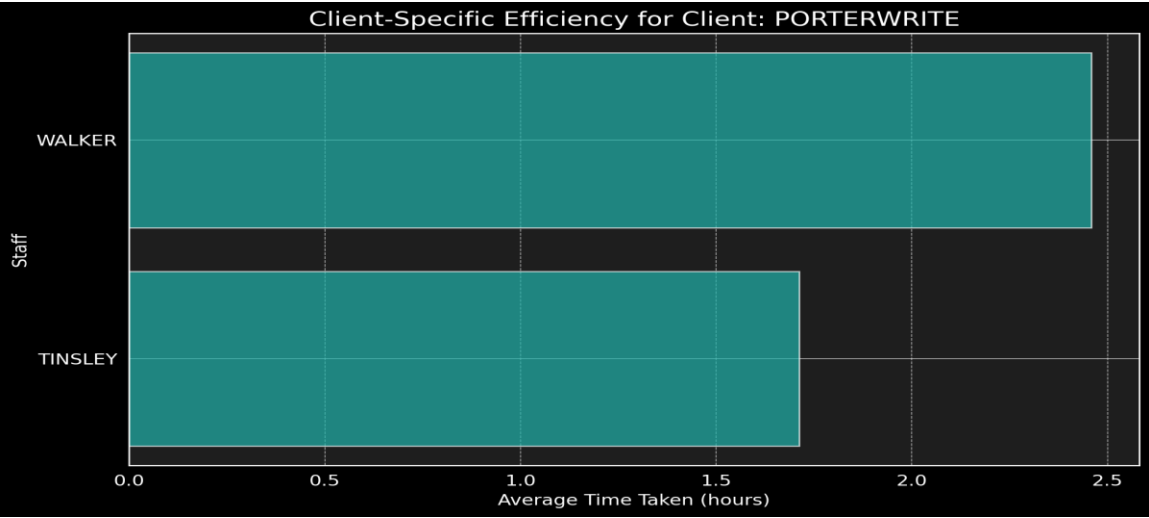


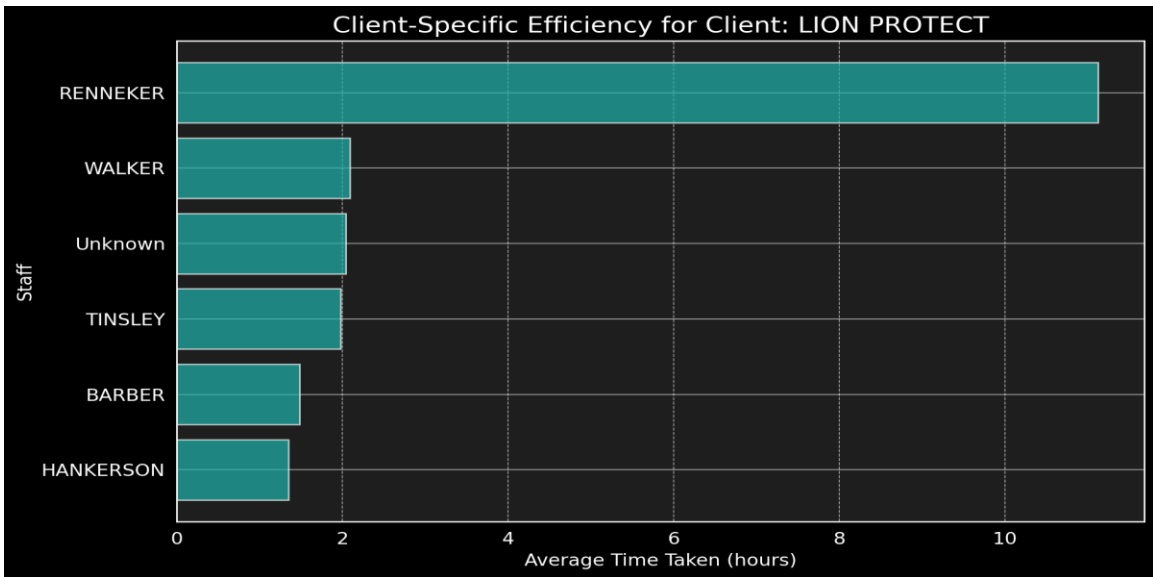


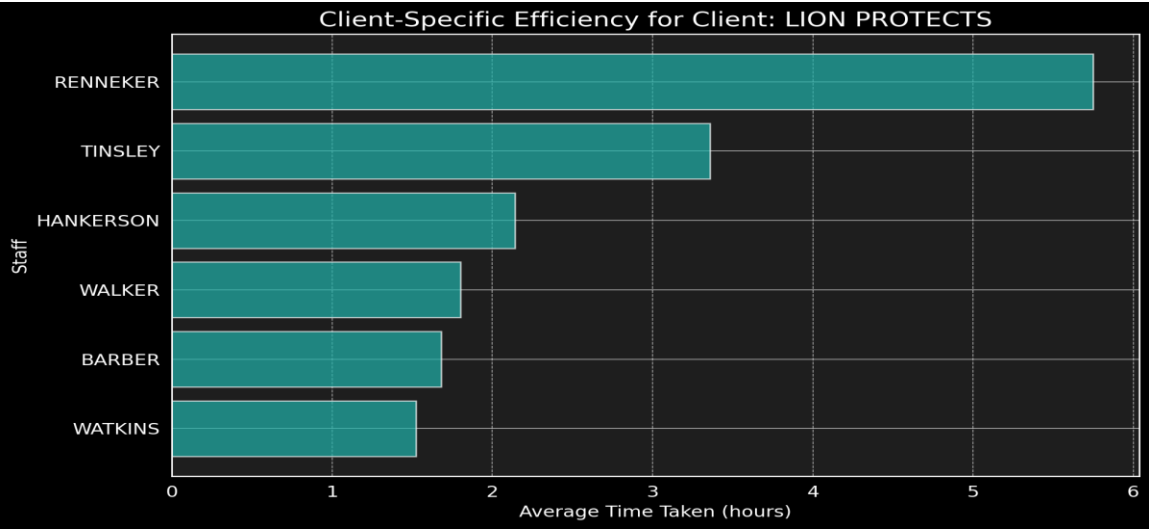


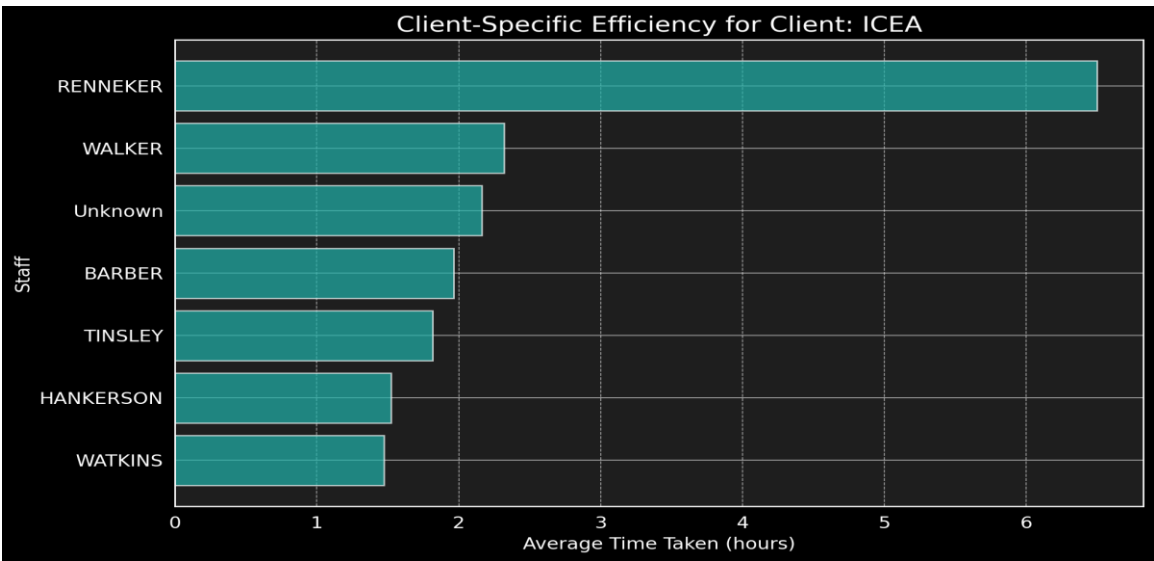


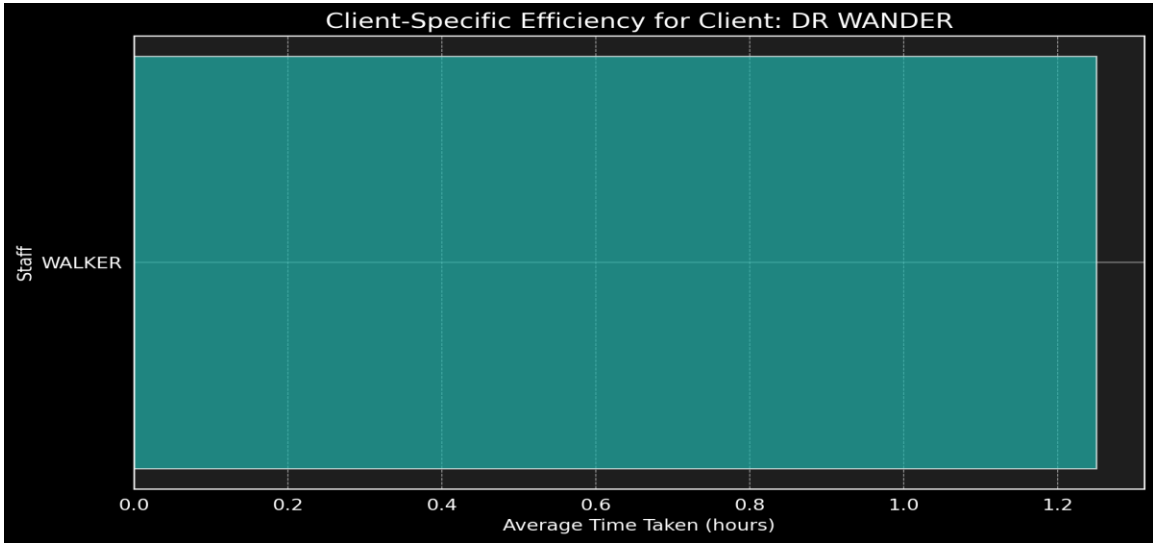
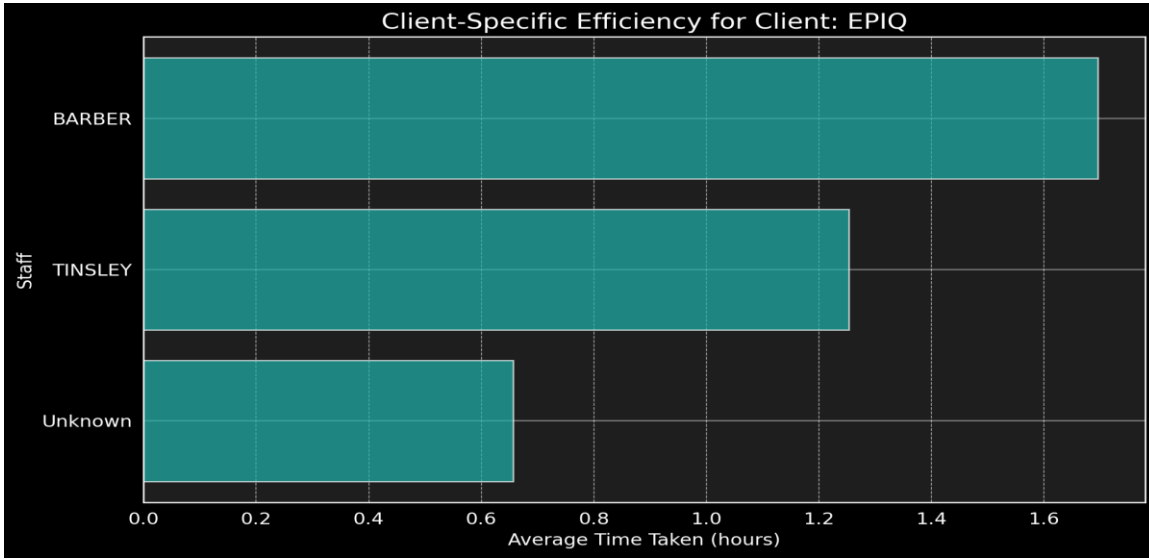
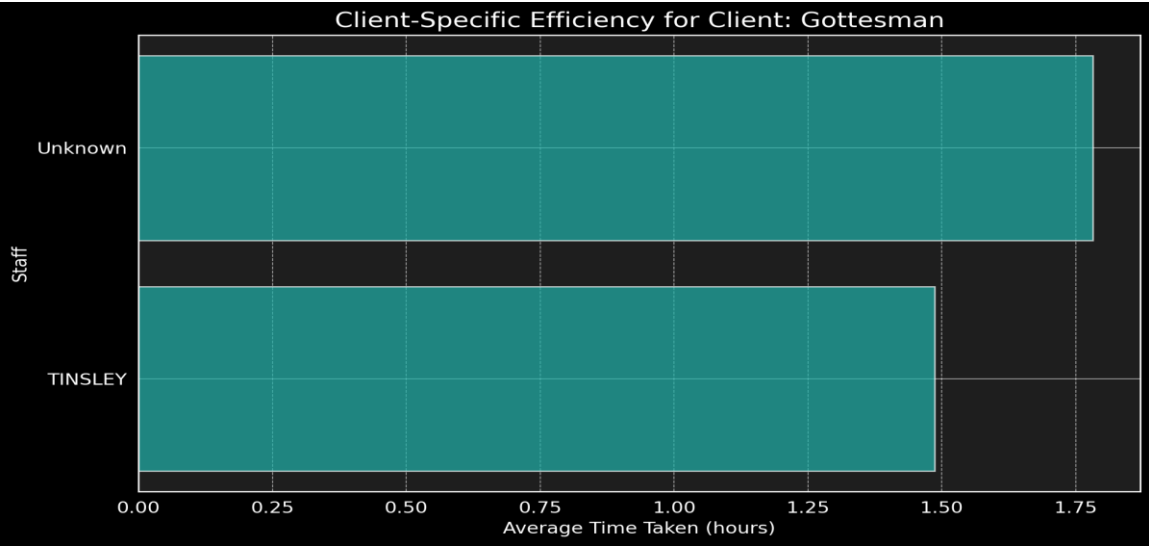


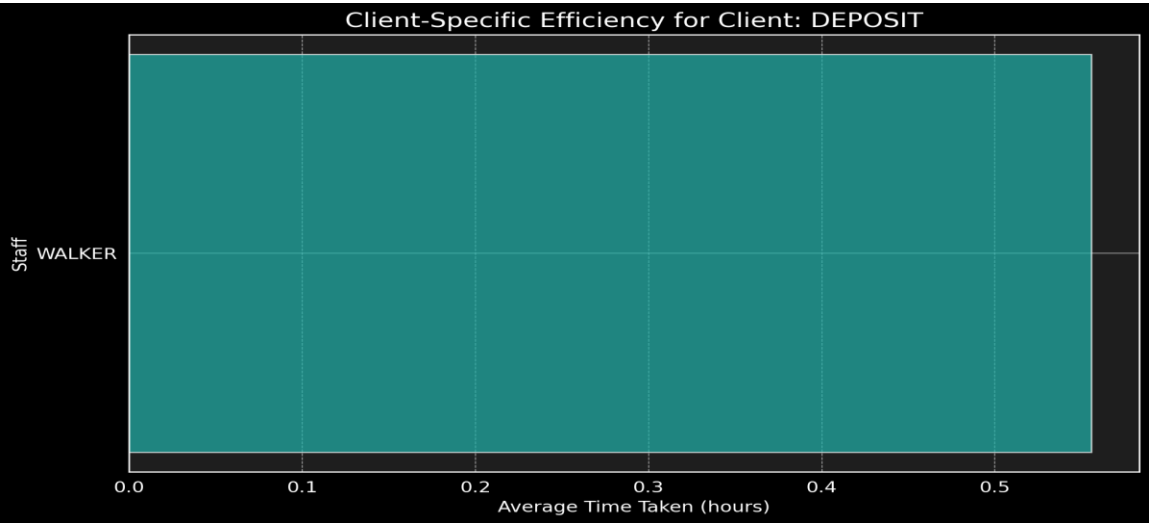
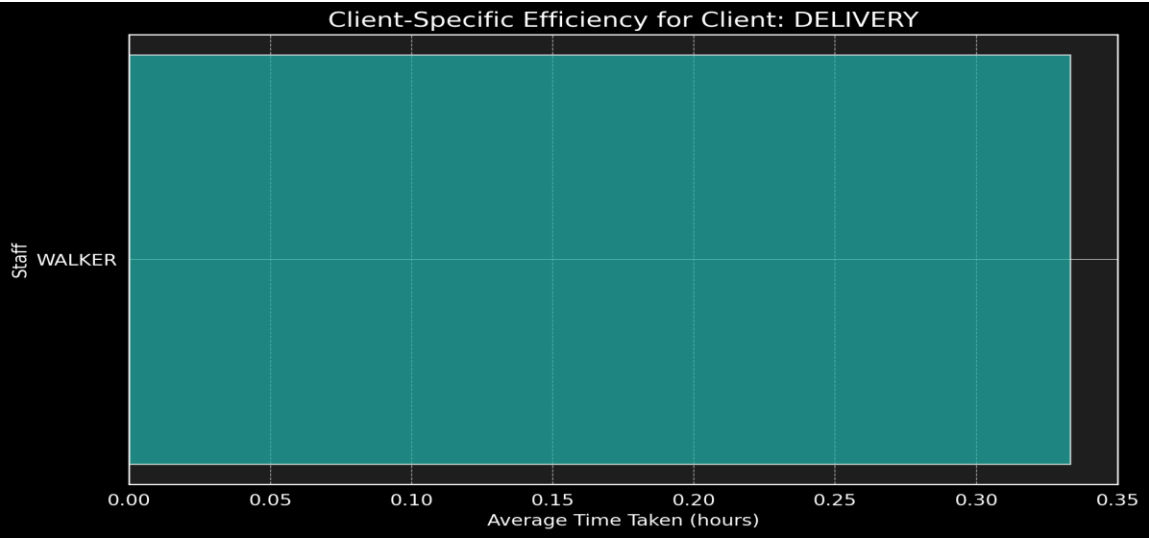




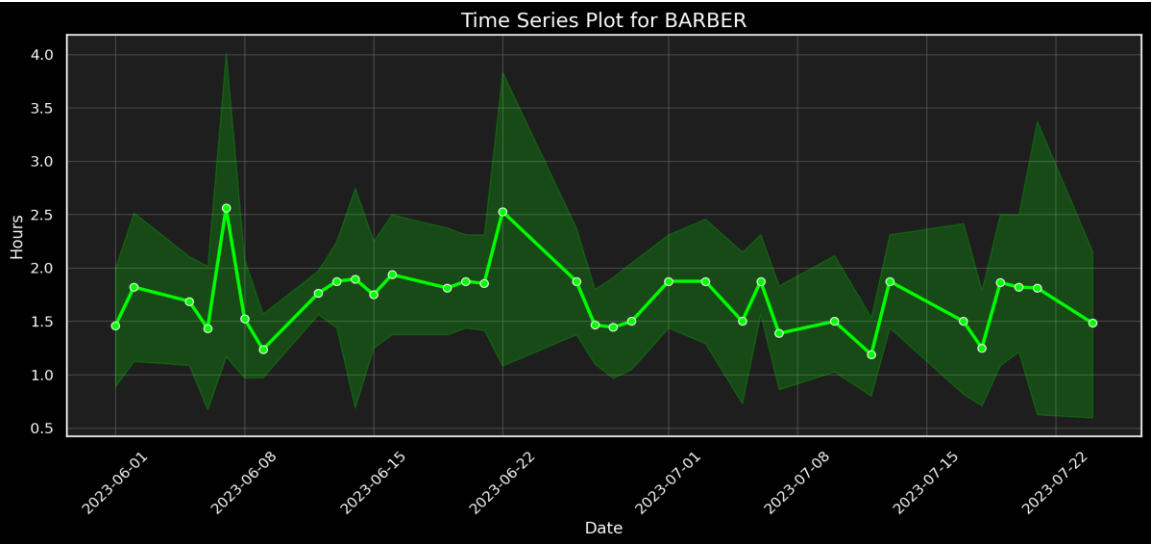
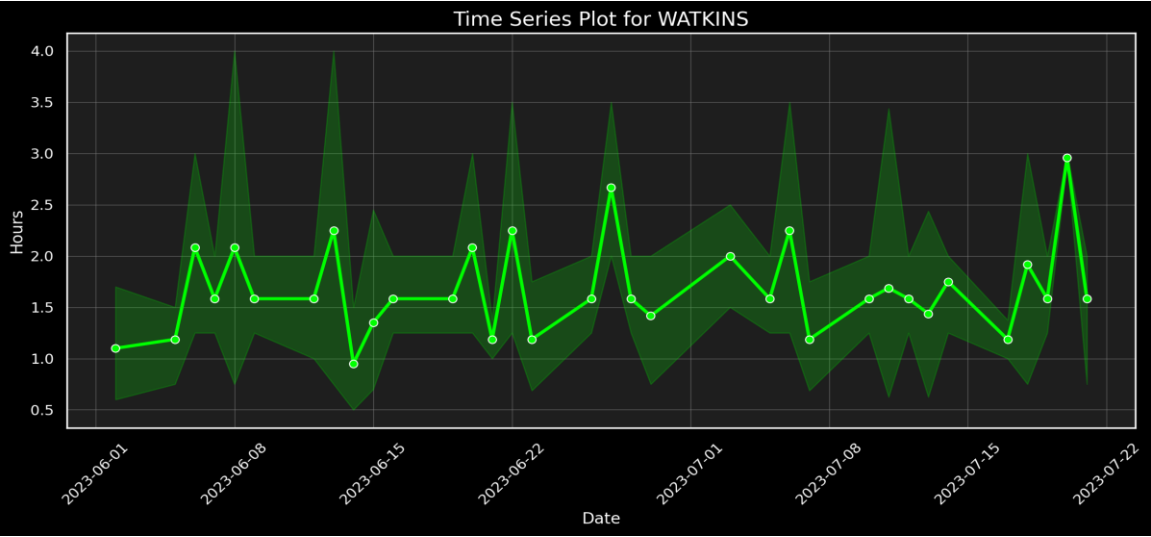
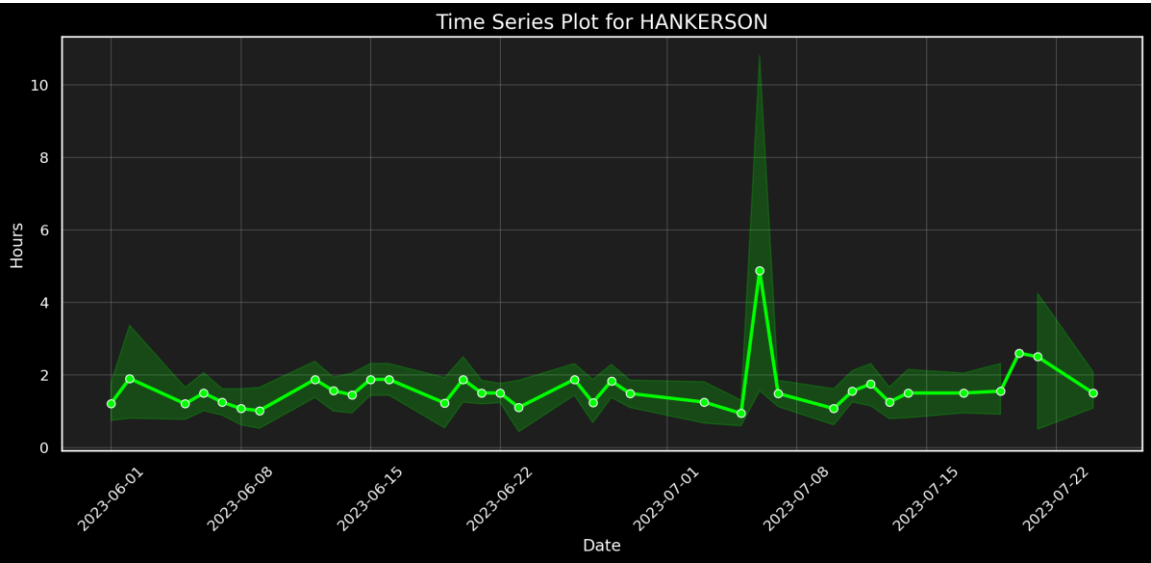


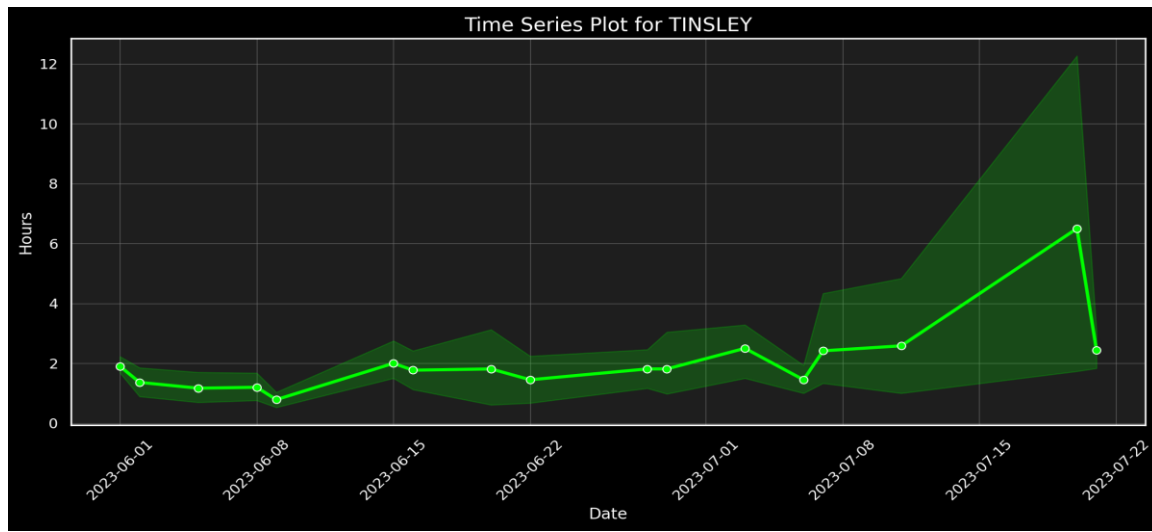
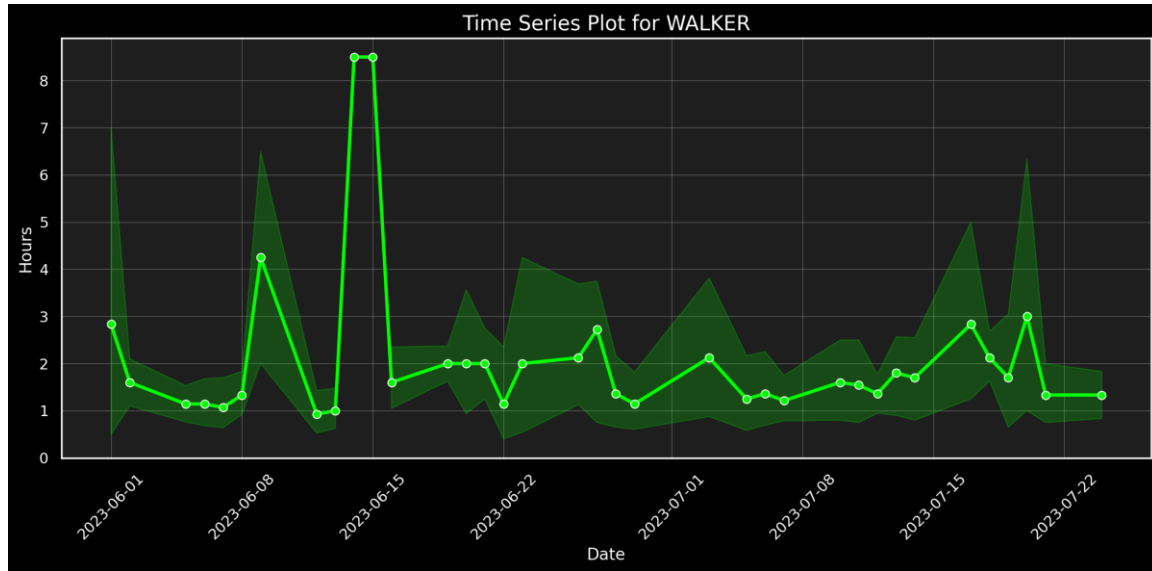
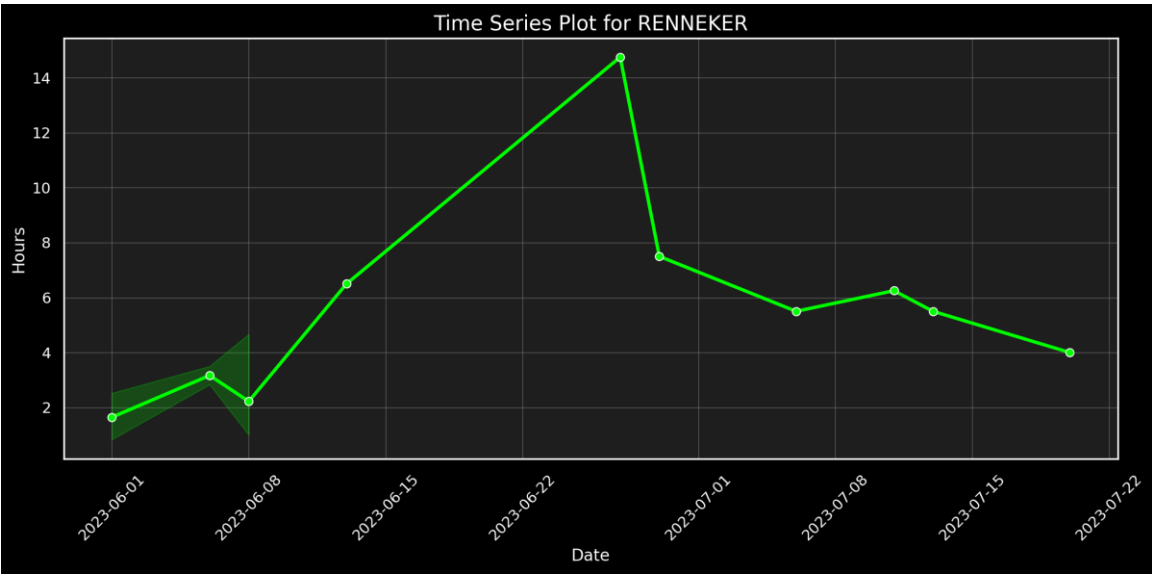


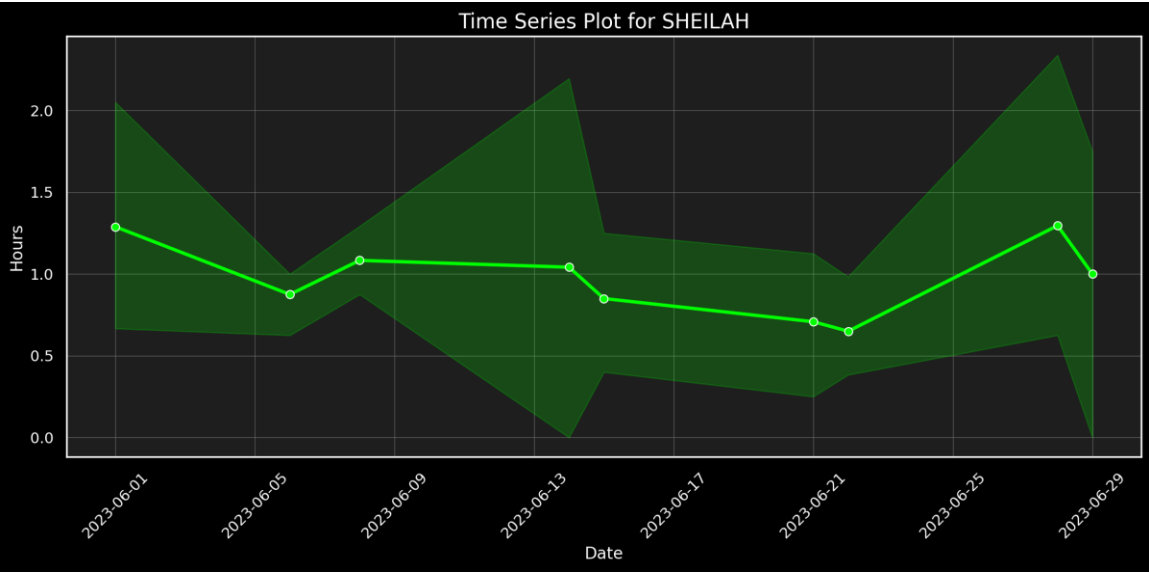
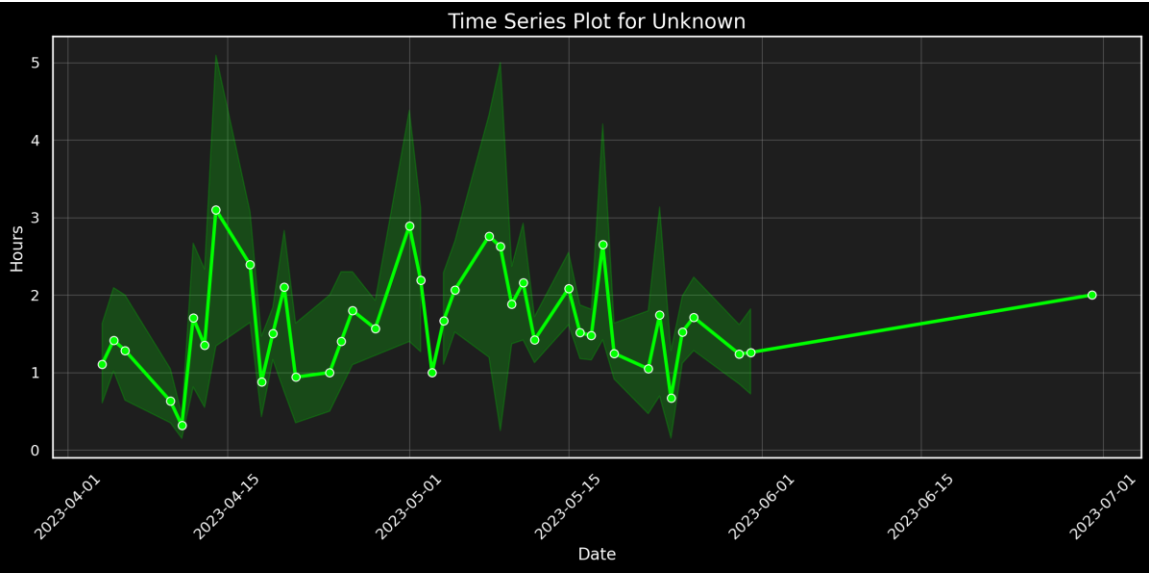












From these results, we can observe that:

1. The maximum time spent on a task in a single day is 14.75 hours by the staff member 'RENEKER' on the 'PREP' activity for the client 'LION PROTECT'.
2. There are several instances where the 'Unknown' staff member has spent more than 6 hours on tasks. These tasks are primarily related to 'US Prod Prep/Scan' and 'US Prod QC' activities.
3. The staff member 'WALKER' has spent 11 hours on a 'COURIER' task for the client 'PRYSMIAN'.
4. The staff member 'TINSLEY' has spent 12.27 hours on a 'QC/REASSEMBLE' task for the client 'LION PROTECTS'.
5. The staff member 'HANKERSON' has spent 13.75 hours on a 'QC' task for the client 'LION PROTECTS'.

Client Name Discrepancies

We can observe several potential issues:

Similar Names: We can see that some client names seem to refer to the same entity but are recorded differently. For example, 'LION PROTECT' and 'LION PROTECTS' or 'LION PROTECTION' might refer to the same client.

Inconsistencies in Name Formats: Some client names have additional information or different formats, such as 'LION PROTECTS/ PHIPPS', 'CHILDREN'S/LINDHURST', and 'CHILDRENS/085', which might indicate inconsistencies in how the client names were recorded.

Possible Typos or Misspellings: Some names, such as 'BONUZZI, LINDHORST' and 'BONEZZI STOCKYARD', might be misspellings or typos.

To clean this data, we could apply some form of text processing or fuzzy matching to group together similar client names.

Conclusion

The analysis of the dataset provided valuable insights into employee efficiency, time management, and task distribution across different projects and clients. However, it's important to note that efficiency is multifaceted, and while time is a crucial aspect, the quality of work, creativity, and teamwork should also be considered when evaluating overall employee performance.

Recommendations

To further enhance the analysis, it's recommended to incorporate additional relevant data, such as task complexity, employee experience level, and client feedback. This would provide a more holistic view of employee efficiency and could help in formulating effective strategies to improve productivity.